Giannina Guzmán Caloca

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Education

University of Maryland

Degree: **M.S., Astronomy** — Graduation Date: January 2023

Degree: Ph.D., Astronomy — Expected Defense: Spring 2027

Villanova University — **GPA: 3.55**

Degree: **Bachelor of Science, *cum laude*** — Graduation Date: May 2019

Major: Astrophysics and Planetary Science

Minor(s): Communication, Physics

Professional Work Experience

★ **NASA Post-baccalaureate** — **September 2019-August 2020**
Developed software for the Planetary Spectrum Generator — *Goddard Space Flight Center*

★ **Software Engineer** — **June 2019-August 2019**
Developed software and helped design telescope automation framework — *Lowell Observatory*

★ **Software Developer** — **August 2018-June 2019**
Developed software for NASA PDART 'sbpy' project — *Villanova University*

★ **NASA GSFC URAA Intern** — **June 2018-August 2018**
Goddard Space Flight Center

★ **Public Observatory Supervisor** — **August 2015-May 2017**
Villanova University

Teaching

★ **NASA Funded Taller Futur@ Astronom@** — **Summer 2024-present**
Teaches a 2.5 hour intro to Astrobiology lecture in Spanish + designs homework

★ **Teaching Assistant** — **August 2020-2023**
ASTR 101 - "Introduction to Astronomy" — *University of Maryland*
ASTR 220 - "Collisions in Space – The Threat of Asteroid Impacts"

★ **Teaching Assistant** — **August 2016-May 2018**
Villanova University

Awards and Honoraries

★ **ExoSLAM 2023 Summer School Attendee** — **Summer 2023**

★ **Jason A. Cardelli Memorial Award for Undergraduate Research** — **Spring 2019**
"The Jason A. Cardelli Memorial Award for Undergraduate Research is presented to a graduating Astronomy & Astrophysics major whose body of undergraduate research work exhibits particularly high standards of independence, originality, and quality." (Villanova University Website)

★ **First place in Villanova's 2018 Sigma Xi CRF Poster Symposium** — **Spring 2018**

Publications

[1] **Guzman Caloca, G**., et al. (2025). GEMS JWST: Hold on to your HATS(-6 b), a sub-solar metallicity planet with water, methane and ammonia in its atmosphere. [in review August 2026]

[2] Kanodia, S., **et al**. (2026). GEMS JWST: A sub-Solar metallicity atmosphere for giant planet TOI-5293Ab orbiting a rapidly changing M-dwarf. DOI: https://doi.org/10.48550/arXiv.2603.16464

[2] Ashtari, R., **et al.** (2025). GEMS JWST: HATS-75 b, A giant planet with a sub-solar metallicity atmosphere orbiting an M-dwarf. DOI: https://doi.org/10.48550/arXiv.2604.07268

[3] Cañas, C. I., **et al.** (2025). GEMS JWST: Transmission spectroscopy of TOI-5205b reveals significant stellar contamination and a metal-poor atmosphere. DOI: doi:10.48550/arXiv.2502.06966

[4] Bell, T., **et al.** (2022). Eureka!: An End-to-End Pipeline for JWST Time-Series Observations. JOSS. DOI: https://doi.org/10.48550/arXiv.2207.03585

[5] van Belle, G.T., Collins, M., **Guzman, G.**, Mommert, M. (2020). Improved ASCOM Dome Following. Research Notes of the AAS. DOI:10.3847/2515-5172/abb29b

[6] **Guzman, G.**, Sion, E., & Godon, P. (2019). FUSE and IUE Spectroscopy of the Prototype Dwarf Nova ERUrsa Majoris during Quiescence. Astronomical Journal. DOI:10.3847/1538-3881/ab322f

[7] Ginsburg, A., **et al.** (2019). astroquery: An Astronomical Web-querying Package in Python. Astronomical Journal. DOI:10.3847/1538-3881/aafc33

[8] Mommert, M., **et al.** (2019). sbpy: A Python module for small-body planetary astronomy. JOSS. DOI: 10.21105/joss.01426

[9] Campbell, H., Sheldon, Z., Gibson, J., **Guzman, G.** (2020). Technological and Mediated Identity in American Multisite Churches. Ecclesial Practices. DOI:10.1163/22144417-bja10002

Presentations

★ **Talk @ Cool Stars 2023 Tokyo, Japan** **June 2026**
Hidden GEMS: The atmospheres of Giant Exoplanets around M-dwarf Stars

★ **Poster @ Cool Stars 2023 Tokyo, Japan** **June 2026**
Hidden GEMS: The atmospheres of Giant Exoplanets around M-dwarf Stars

★ **ESPF Seminar @ STScI** **May 2026**
Of sub-Neptunes and GEMS: A Broad Look into Atmospheres with No Solar System Analogs

★ **Carnegie EPL Astronomy Seminar Series** **April 2026**
Of sub-Neptunes and GEMS: A Broad Look into Atmospheres with No Solar System Analogs

★ **Exoplanet Seminar @ Goddard Space Flight Center** **March 2026**
Of sub-Neptunes and GEMS: A Broad Look into Atmospheres with No Solar System Analogs

★ **Planetary Lunch @ UC Santa Cruz** **March 2026**
Of sub-Neptunes and GEMS: A Broad Look into Atmospheres with No Solar System Analogs

★ **ERES X Presenter** **June 2025**
Looking at Giant Exoplanets around M-dwarfs (GEMS) with JWST: NIRSpec Transmission Spectroscopy of the Warm Saturn HATS-6 b

★ **Looking for GEMS** **February 2025**
Talk for the AstroAnimation Collaboration @ Maryland Institute College of Art

★ **AAS 245th Meeting Talk** **January 2025**
Looking at Giant Exoplanets around M-dwarfs (GEMS) with JWST: NIRSpec Transmission Spectroscopy of the Warm Saturn HATS-6 b

★ **AAS 245th Meeting Talk** **January 2025**
Being Stardust: A Collaboration between Arts and Sciences

★ **Early Career Science Forum Talk** **October 2024**
@ NASA GSFC - Looking at Giant Exoplanets around M-dwarfs (GEMS)

★ **Eureka! An End-to-End Pipeline for JWST Time Series Observations** **December 2023**
ExoVAST (Virtual Astronomy Talks) seminar talk

★ **Gridtrievals? A Comparative Study of Retrieval Techniques** **June 2023**
ExoClimes 2023 Poster

★ **Eureka! An End-to-End Pipeline for JWST Time Series Observations** **February 2023**
First EMAC Workshop on Open-Access Exoplanet Modeling & Analysis Tools

★ **Accurate (Exo)planetary Retrievals via High Dimensional Bayesian Samplers** **August 2020**
NASA GSFC Planetary Science Division Symposium

★ **The Red Thumbs: Growing Plants on Martian Regolith Simulant** **January 2019**
AAS Meeting #233 Poster - Education Category

★ **Designing a Python Module for the Calculation of [..] in Comets** **January 2019**
AAS Meeting #233 Poster

★ **Digitizing Villanova University's Eclipsing Binary Card Catalogue** **January 2018**
AAS Meeting #231 Poster

Community Service

★ **Space Sciences Outreach Cooperative Co-founder and Co-chair** **May 2023-Present**
Co-founder and co-chair of the University of Maryland space sciences outreach coop (SSOC) + lead in several outreach activities throughout each year

★ **Class Representative** **January 2021-Present**
Graduate Council, University of Maryland Department of Astronomy

★ **Executive Secretary** **May 2026**
NASA review panel

★ **Executive Secretary** **April 2026**
NASA review panel

★ **Executive Secretary** **December 2025**
NASA review panel

★ **NASA JWST x Minecraft Education Collaboration Interviewee** **September 2024**
Presenter for a NASA JWST x Minecraft Education collaboration

★ **Being Stardust Workshop @ Library of Congress** **April 2024**
Participant in an art meets astrobiology collaboration outreach workshop

★ **Prospective Visit Organizer** **Fall 2022-Spring 2024**
Organized the graduate prospective visit for the Department of Astronomy

★ **Workshops in Applying to Astronomy Graduate School co-lead** **Fall 2023**
Helped establish and lead a workshop series at UMD for astronomy graduate school applications directed at local undergraduates

★ **#MathGals Special Guest** **Fall 2022**
Invited special guest for Pasadena ISD school district #mathgals club

★ **UMD Astronomy EDI Committee Member** **Fall 2021-Fall 2024**

★ **UMD Astronomy Undergraduate Mentorship Program Mentor** **Fall 2021-Fall 2023**
Mentor for undergraduate students, helps with professional development

★ **Mental Health Task Force** **Spring 2021**
Co-authored the mental health report for UMD's Astronomy Department; based on a dedicated mental health survey done in 2020

★ **AbGradCon2022 Organizing Committee** **Fall 2021-Fall 2022**
Position of leadership: Public and Media Relations

★ **Executive Secretary** **April 2020**
NASA Review Panel

★ **GSMI Cientifico Latino Volunteer** **August 2020-May 2021**
Student Mentor

★ **The Superlative** **January 2016-May 2019**
Position of leadership (2016-2019): Public Relations and media representative

★ **Villanova Astronomical Society** **August 2015-May 2019**
Position of leadership (2017-2018): Treasurer

★ **All Hands-On Science** **May 2017-May 2018**
Volunteering

Awarded Observation Proposals

★ **Target lead:** JWST (Cycle 2 GO #3171; 132.39 hrs)
Red Dwarfs and the Seven Giants: First Insights into the Atmospheres of Giant Exoplanets around M-dwarf Stars

★ **Target lead:** HST (Cycle 30 #17192; 116 orbits)
The SPACE Program: a Sub-neptune Planetary Atmosphere Characterization Experiment

★ **Co-I:** JWST (Cycle 4 GO #7683; 22.6 hrs)
A Parched Giant orbiting a Red Dwarf: Fact or Fiction?

★ **Co-I:** JWST (Cycle 3 GO #6284; 13.68 hrs)
Searching for Signatures of Surface-Atmosphere Interaction on a Small Planet in its Magma Era

★ **Co-I:** JWST (Cycle 3 GO #5959; 129.96 hrs)
KRONOS: Keys to Revealing the Origin and Nature Of sub-neptune Systems